

mTORC1 signaling: what we still don't know.

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The mammalian target of rapamycin (mTOR) is a protein kinase that plays key roles in cellular regulation. It forms complexes with additional proteins. The best-understood one is mTOR complex 1 (mTORC1). The regulation and cellular functions of mTORC1 have been the subjects of intense study; despite this, many questions remain to be answered. They include questions about the actual mechanisms by which mTORC1 signaling is stimulated by hormones and growth factors, which involves the small GTPase Rheb, and by amino acids, which involves other GTPase proteins. The control of Rheb and the mechanism by which it activates mTORC1 remain incompletely understood. Although it has been known for many years that rapamycin interferes with some functions of mTORC1, it is not known how it does this, or why only some functions of mTORC1 are affected. mTORC1 regulates diverse cellular functions. Several mTORC1 substrates are now known, although in several cases their physiological roles are poorly or incompletely understood. In the case of several processes, although it is clear that they are regulated by mTORC1, it is not known how mTORC1 does this. Lastly, mTORC1 is implicated in ageing, but again it is unclear what mechanisms account for this. Given the importance of mTORC1 signaling both for cellular functions and in human disease, it is a high priority to gain further insights into the control of mTORC1 signaling and the mechanisms by which it controls cellular functions and animal physiology.